

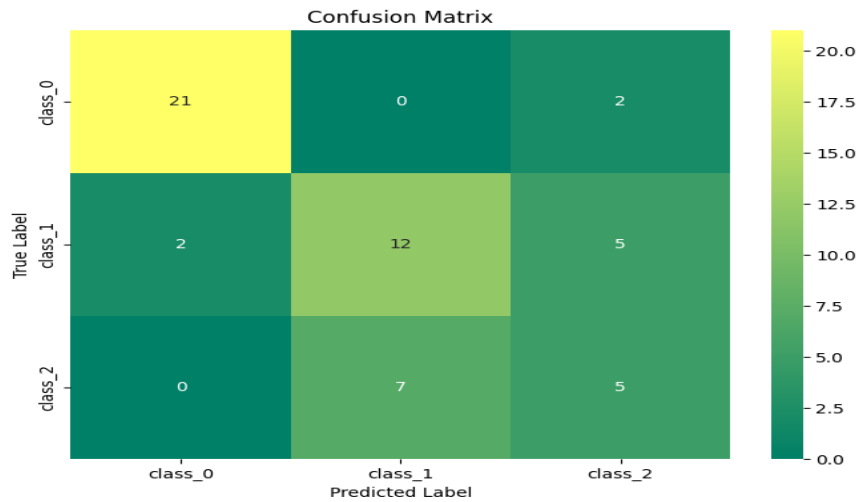
EXP NO 6:

```
[12]
✓ 1s
import pandas as pd
import pandas as pd
import matplotlib.pyplot as plt

import seaborn as sns
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

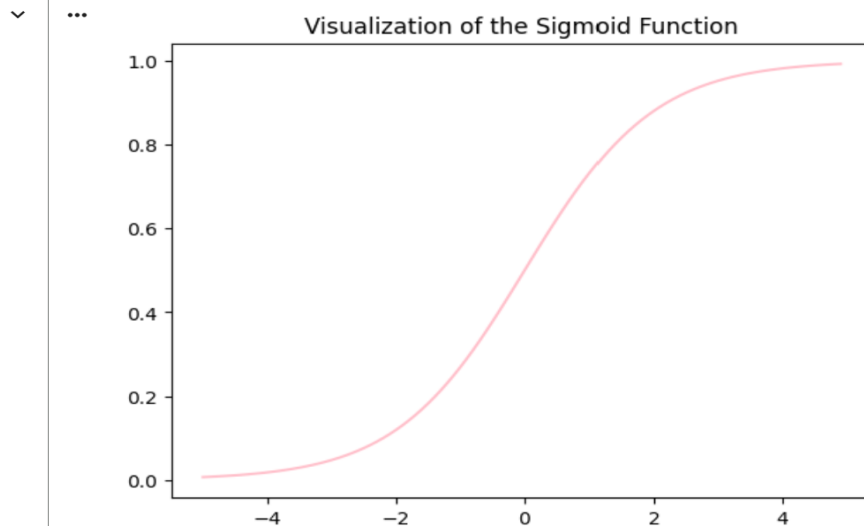
wine=load_wine()
data=pd.DataFrame(data=wine.data,columns=wine.feature_names)
data['Target']=wine.target
x=data.drop('Target',axis=1)
y=data['Target']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=1)
model=KNeighborsClassifier(n_neighbors=5)
model.fit(x_train,y_train)
y_pred=model.predict(x_test)
accuracy=accuracy_score(y_test,y_pred)
print("accuracy:",accuracy)
conf_matrix = confusion_matrix(y_test,y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='summer',
            xticklabels=wine.target_names, yticklabels=wine.target_names)
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix')
plt.show()
```

... accuracy: 0.7037037037037037



EXP NO 7

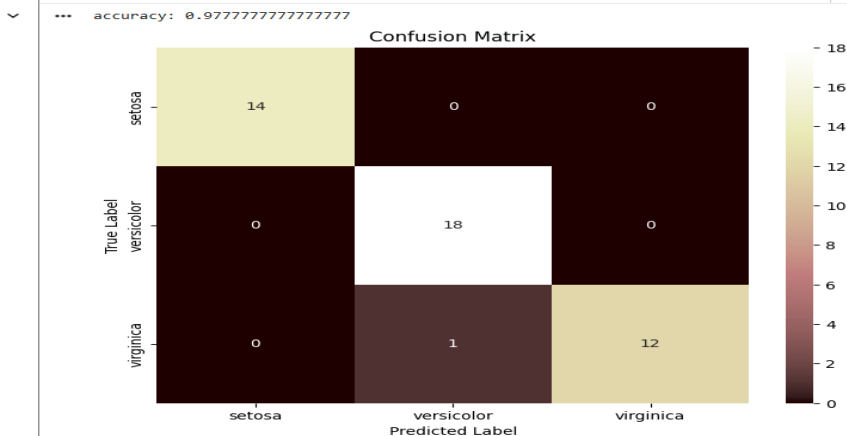
```
[14]
✓ Os
import numpy as np
import matplotlib.pyplot as plt
def sigmoid(z):
    return 1 / (1 + np.exp(-z))
plt.plot(np.arange(-5, 5, 0.1), sigmoid(np.arange(-5, 5, 0.1)),color='pink')
plt.title('Visualization of the Sigmoid Function')
plt.show()
```



EXP NO 8

```
[15]
✓ Os
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
iris=load_iris()
data=pd.DataFrame(data=iris.data,columns=iris.feature_names)
data['Species']=iris.target
x=data.drop('Species',axis=1)
y=data['Species']

x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=1)
model=KNeighborsClassifier(n_neighbors=5)
model.fit(x_train,y_train)
y_pred=model.predict(x_test)
accuracy=accuracy_score(y_test,y_pred)
print("accuracy:",accuracy)
conf_matrix = confusion_matrix(y_test,y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='pink',
            xticklabels=iris.target_names, yticklabels=iris.target_names)
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix')
plt.show()
```



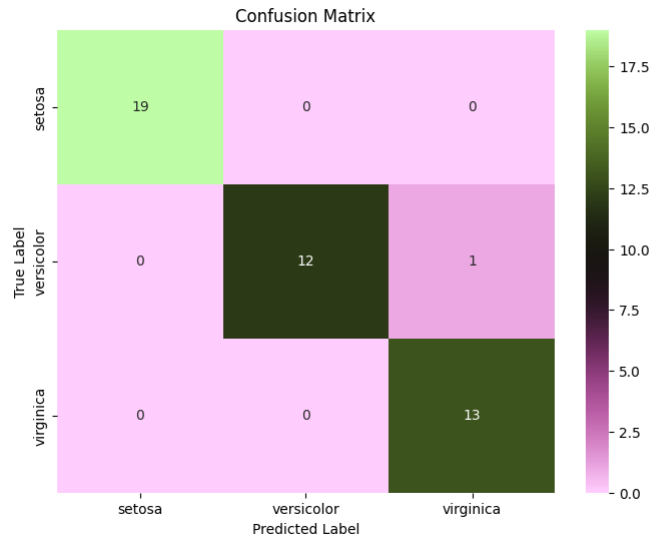
EXP NO 9

```
#naive bayes iris
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix

iris = load_iris()

data = pd.DataFrame(data=iris.data, columns=iris.feature_names)
data['species'] = iris.target
X = data.drop('species', axis=1)
y = data['species']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
                                                    random_state=42)
model = GaussianNB()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("accuracy:", accuracy)
conf_matrix = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='magma',
            xticklabels=iris.target_names, yticklabels=iris.target_names)
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix')
plt.show()
```

... accuracy: 0.9777777777777777



EXP NO 10

```
) import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
iris = load_iris()
data = pd.DataFrame(data=iris.data, columns=iris.feature_names)

data['species'] = iris.target
X = data.drop('species', axis=1)
y = data['species']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
                                                    random_state=1)
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("accuracy:", accuracy)
conf_matrix = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='berlin',
            xticklabels=iris.target_names, yticklabels=iris.target_names)
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix')
plt.show()
```

accuracy: 0.9666666666666667

